ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2024 P2HR Worked Solutions

Jun 2024 | P2HR

33 questions ▪ question image followed by worked solution

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STUDENT EDITION
Worked answers included

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PAST PAPER QUESTION

Q#: 1 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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1 Write 1400 as a product of powers of its prime factors.
Show your working clearly.

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Prime Factors, HCF & LCM**

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WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$1400 = 14 \times 100$$

$$1400 = (2 \times 7)(2^2 \times 5^2)$$

$$1400 = 2^3 \times 5^2 \times 7$$

FINAL ANSWER

$2^3 \times 5^2 \times 7$

PAST PAPER QUESTION

Q#: 2 **Marks: 4**

TOPIC: **Transformations**

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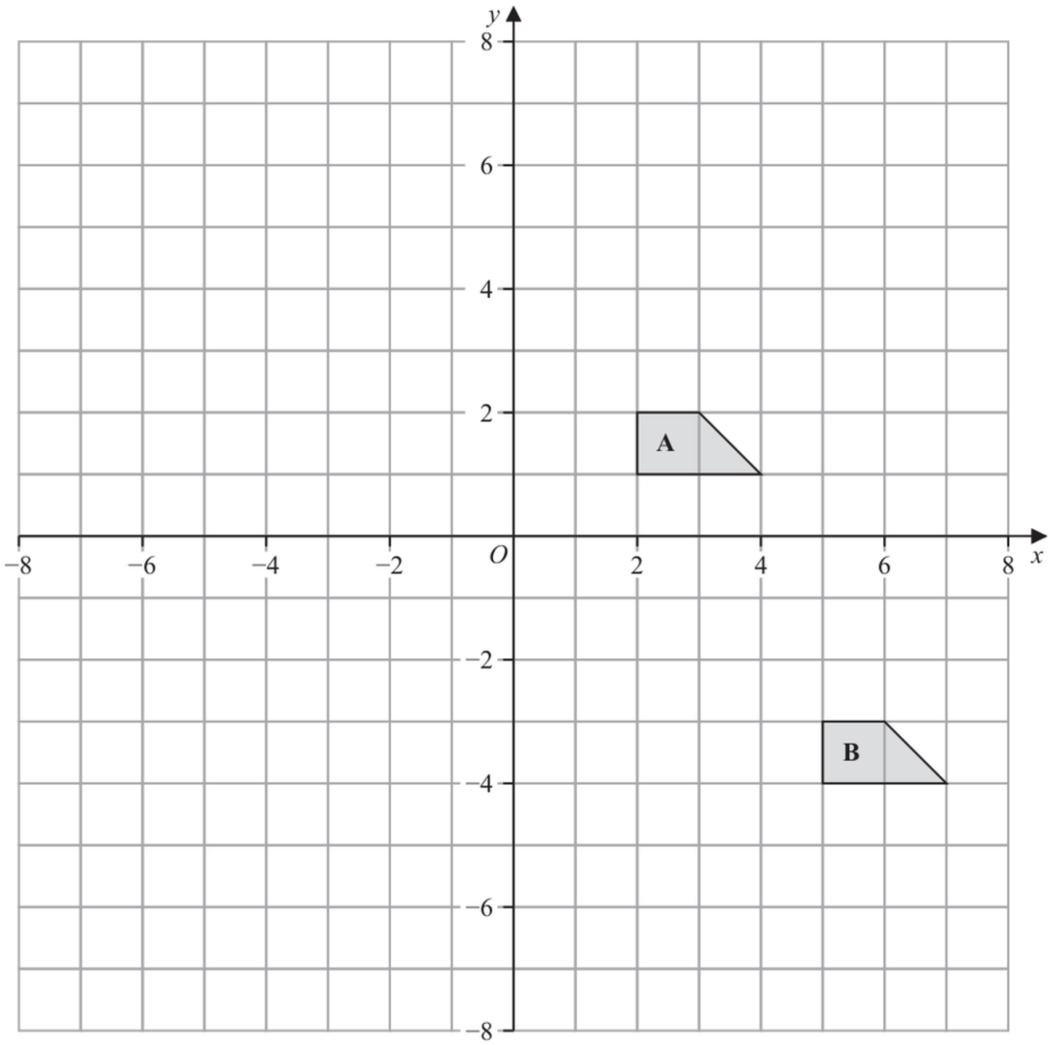
2



(a) Describe fully the single transformation that maps shape **A** onto shape **B**

....

....

(2)

(b) On the grid above, rotate shape **A** 180° about (−1, 0)
Label your shape **C**

(2)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION Q#: 2 Marks: 4

TOPIC: **Transformations**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

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Topic check. Transformations. The tag is correct.

Solution

(a) Shape A moves to shape B by the vector

$$\begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

So the transformation is a translation by $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$.

(b) For a 180° rotation about $(-1, 0)$,

$$(x, y) \mapsto (-2 - x,\ -y)$$

The image has vertices

$$(-4, -1),\ (-6, -1),\ (-5, -2),\ (-4, -2)$$

FINAL ANSWER

(a) translation by $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$; (b) vertices $(-4, -1), (-6, -1), (-5, -2), (-4, -2)$.

PAST PAPER QUESTION

Q#: 3 Marks: 2

TOPIC: **Statistics Toolkit**

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3 Here is a list of four numbers written in ascending order of size

$$x \qquad x \qquad y \qquad 15$$

where x and y are integers.

The numbers have

a median of 12.5
a range of 4

Find the value of x and the value of y

$x =$ ....

$y =$ ....

(Total for Question 3 is 2 marks)

PAST PAPER SOLUTION

Q#: 3 Marks: 2

TOPIC: **Statistics Toolkit**

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The range is 4:

$$15 - x = 4$$

$$x = 11$$

The median is 12.5:

$$\frac{x + y}{2} = 12.5$$

$$x + y = 25$$

$$y = 25 - 11 = 14$$

FINAL ANSWER

$x = 11, \ y = 14.$

PAST PAPER QUESTION Q#: 4 Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

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4 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
$A = \{\text{factors of } 6\}$
$B = \{\text{prime numbers}\}$

(a) List the members of the set

(i) $A \cup B$

....

(1)

(ii) A'

....

(1)

Harpreet states that $A \cap B = \varnothing$

Harpreet is incorrect.

(b) Explain why.

....

(1)

C is a set with 4 members such that

the set $A \cap C$ has 2 members
the set $B \cap C$ has 2 members

Set $A \cap C$ and set $B \cap C$ have no members in common.

(c) List the 4 members of set C

....

(2)

(Total for Question 4 is 5 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 5

TOPIC: **Set Notation & Venn Diagrams**

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WORKED SOLUTION

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Topic check. Set notation and Venn diagrams. The tag is correct.

Method

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 2, 3, 6\}$$

$$B = \{2, 3, 5, 7\}$$

So

$$A \cup B = \{1, 2, 3, 5, 6, 7\}$$

and

$$A' = \{4, 5, 7, 8, 9, 10\}$$

Harpreet is not correct because 2 and 3 are in both A and B.

$$A \cap B = \{2, 3\}$$

For C, it has 4 members, $n(A \cap C) = 2$, $n(B \cap C) = 2$, and no member is in both of those overlaps. So choose the two elements of A only and the two elements of B only:

$$C = \{1, 5, 6, 7\}$$

FINAL ANSWER

(a)(i) $\{1, 2, 3, 5, 6, 7\}$, (a)(ii) $\{4, 5, 7, 8, 9, 10\}$, (b) Harpreet is incorrect, (c) $C = \{1, 5, 6, 7\}$

PAST PAPER QUESTION Q#: 5 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

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5 The diagram shows the design for a badge, which will be made using wire.
The design is a circle inside a square *ABCD*

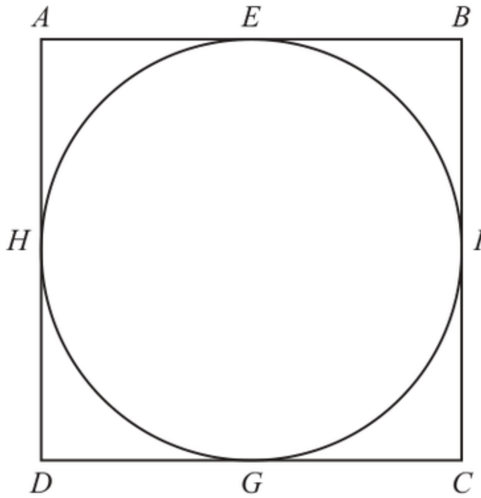

Diagram **NOT** accurately drawn

The circle touches the square at the points *E*, *F*, *G* and *H*

The area of the square is 81 cm^2

Calculate the total length of wire that will be needed to make the square and the circle.
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

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WORKED SOLUTION

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Topic check. Corrected to Circles, Arcs & Sectors.

Solution

The square has area 81 cm^2, so its side length is

$$\sqrt{81} = 9 \text{ cm}$$

Perimeter of the square:

$$4 \times 9 = 36$$

The circle has diameter 9 cm, so its circumference is

$$\pi d = 9\pi$$

Total wire length:

$$36 + 9\pi = 64.274\ldots$$

FINAL ANSWER

64.3 cm to 3 significant figures.

PAST PAPER QUESTION

Q#: 6 Marks: 8

TOPIC: **Solving Linear Equations**

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6 (a) Solve $\dfrac{2f}{3} = 4f - 17$

Show clear algebraic working.

$f =$ ....

(3)

(b) Simplify $(e + 12)^0$ where $e > 0$

....

(1)

(c) Simplify fully $\dfrac{12a^4h^6}{4ah^2}$

....

(2)

(d) Factorise fully $20x^5y + 12x^3y^4$

....

(2)

(Total for Question 6 is 8 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 8

TOPIC: **Solving Linear Equations**

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WORKED SOLUTION

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Topic check. Corrected to solving linear equations. The original tag was factorising, but the equation is the largest single part.

Solution

(a)

$$\frac{2f}{3} = 4f - 17$$

$$2f = 12f - 51$$

$$10f = 51$$

$$f = \frac{51}{10}$$

(b) $(e + 12)^0 = 1$.

(c)

$$\frac{12a^4h^6}{4ah^2} = 3a^3h^4$$

(d)

$$20x^5y + 12x^3y^4 = 4x^3y(5x^2 + 3y^3)$$

FINAL ANSWER

$f = \frac{51}{10}$, 1, $3a^3h^4$, $4x^3y(5x^2 + 3y^3)$

PAST PAPER QUESTION

Topic: **Algebraic Roots & Indices**

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Q#: 7 Marks: 2

7 $\dfrac{3^{-2} \times 3^{5}}{3^{10}} = 3^{n}$

Find the value of n

$n =$ ....

(Total for Question 7 is 2 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 2

TOPIC: **Algebraic Roots & Indices**

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Topic check. Algebraic Roots and Indices. The tag is correct.

Method

$$\frac{3^{-2} \times 3^{5}}{3^{10}} = 3^{n}$$

Use index laws:

$$\frac{3^{-2+5}}{3^{10}} = 3^{3-10} = 3^{-7}$$

So

$$n = -7$$

FINAL ANSWER

$n = -7$.

PAST PAPER QUESTION Q#: 8 Marks: 3

TOPIC: **Percentages**

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8 In a sale, all normal prices are reduced by 17%

The sale price of a fridge is 6225 rupees.

Work out the normal price of the fridge.

.... rupees

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 3

TOPIC: **Percentages**

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WORKED SOLUTION

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Topic check. Percentages. The tag is correct.

Method

A reduction of 17% means the sale price is 83% of the normal price.

$$0.83 \times \text{normal price} = 6225$$

$$\text{normal price} = \frac{6225}{0.83} = 7500$$

FINAL ANSWER

7500 rupees

PAST PAPER QUESTION

Q#: 9 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Jun 2024 P2HR - Jun 2024 | P2HR

9 (a) Write 6.04×10^5 as an ordinary number.

....

(1)

(b) Write 0.000 07 in standard form.

....

(1)

(c) Work out $\dfrac{7.6 \times 10^{10}}{4 \times 10^5 - 2 \times 10^4}$

Give your answer in standard form.

....

(2)

(Total for Question 9 is 4 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

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WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$6.04 \times 10^5 = 604000$$

$$0.00007 = 7 \times 10^{-5}$$

For part (c), first simplify the denominator:

$$4 \times 10^5 - 2 \times 10^4 = 400000 - 20000 = 380000$$

$$380000 = 3.8 \times 10^5$$

Then

$$\frac{7.6 \times 10^{10}}{3.8 \times 10^5} = 2 \times 10^5$$

FINAL ANSWER

(a) 604000, (b) 7×10^{-5}, (c) 2×10^5

PAST PAPER QUESTION

Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

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10 The diagram shows a hexagon *ABCDEF*

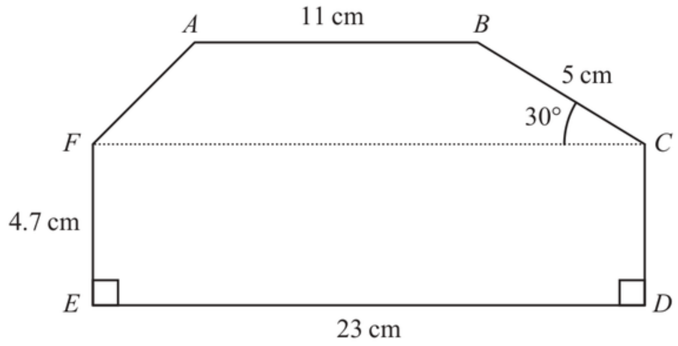

Diagram **NOT** accurately drawn

Angle $BCF = 30°$
AB, FC and ED are parallel.

Calculate the area of *ABCDEF*
Show your working clearly.

.... cm^2

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 5

TOPIC: **Area & Perimeter**

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WORKED SOLUTION

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Topic check. Area & Perimeter. The tag is correct.

Solution

The rectangle $EDCF$ has area

$$23 \times 4.7 = 108.1$$

The height of the top trapezium is

$$5 \sin 30^\circ = 2.5$$

Area of the top trapezium:

$$\frac{1}{2}(11 + 23) \times 2.5 = 42.5$$

Total area:

$$108.1 + 42.5 = 150.6$$

FINAL ANSWER

150.6 cm^2.

PAST PAPER QUESTION

Q#: 11 **Marks: 7**

TOPIC: **Cumulative Frequency Diagrams**

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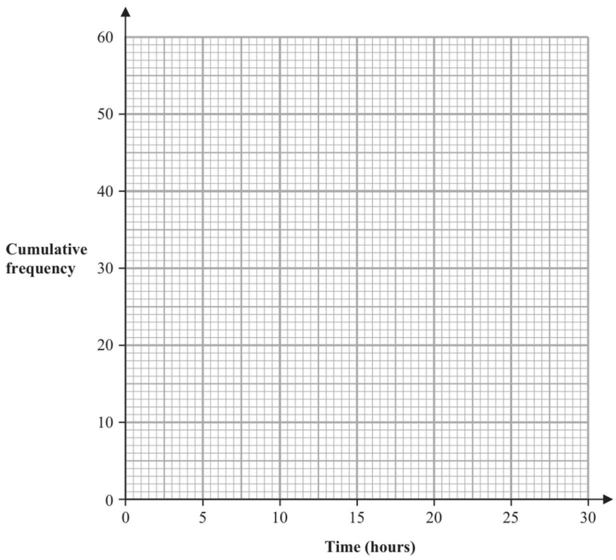
11 The cumulative frequency table gives information about the time, in hours, that each of 60 workers spent working from home in one week.

Time (t hours)	Cumulative frequency
$0 < t \leqslant 5$	6
$0 < t \leqslant 10$	17
$0 < t \leqslant 15$	27
$0 < t \leqslant 20$	42
$0 < t \leqslant 25$	53
$0 < t \leqslant 30$	60

(a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

(b) Use your graph to find an estimate for the interquartile range of the times.

.... hours
(2)

25 workers spent more than W hours working from home.

(c) Use your graph to find an estimate for the value of W

$W =$ ....
(2)

One of the 60 workers is chosen at random.
This worker spent H hours working from home.

(d) Find the probability that $5 < H \leqslant 10$

....
(1)

(Total for Question 11 is 7 marks)

PAST PAPER SOLUTION Q#: 11 Marks: 7

TOPIC: **Cumulative Frequency Diagrams**

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WORKED SOLUTION

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Topic check. Cumulative Frequency Diagrams. The tag is correct.

Solution

The cumulative frequencies are

$$6,\ 17,\ 27,\ 42,\ 53,\ 60$$

Plot

$$(0,0),(5,6),(10,17),(15,27),(20,42),(25,53),(30,60)$$

For 60 workers,

$$Q_1 = 15, \qquad Q_3 = 45$$

From the graph,

$$Q_1 \approx 9.1, \qquad Q_3 \approx 21.4$$

$$\text{IQR} \approx 21.4 - 9.1 = 12.3$$

If 25 workers did more than W hours, then 35 workers did W hours or less.
From the graph, $W \approx 17.7$.

$$P(5 < H \leq 10) = \frac{17-6}{60} = \frac{11}{60}$$

FINAL ANSWER
IQR about 12.3, $W \approx 17.7$, and $\frac{11}{60}$.

PAST PAPER QUESTION Q#: 12 Marks: 4

Topic: **Congruence, Similarity & Geometrical Proof**

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12

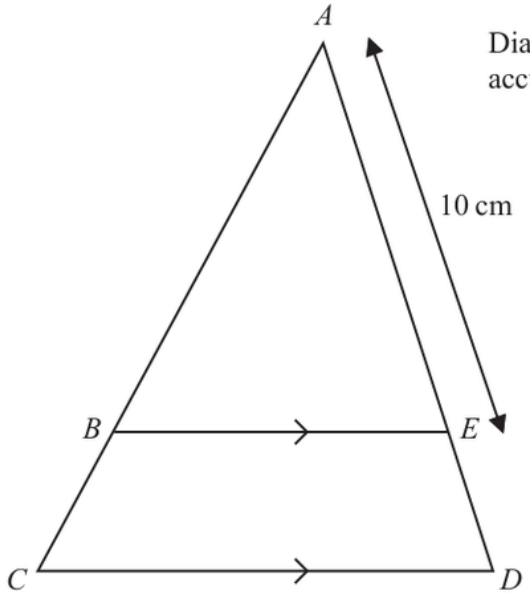

Diagram **NOT** accurately drawn

In the diagram, *ABC* and *AED* are straight lines.
BE is parallel to *CD*

$AE = 10\,\text{cm}$ and $CD = 1.5 \times BE$

(a) Work out the length of *ED*

.... cm

(2)

$AB = (2x + 5)\,\text{cm}$ and $BC = (3x - 5)\,\text{cm}$

(b) Work out the value of x

$x =$ ....

(2)

(Total for Question 12 is 4 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 4

Topic: **Congruence, Similarity & Geometrical Proof**

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WORKED SOLUTION

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Topic check. Corrected to congruence/similarity. The parallel line gives similar triangles.

Solution

Since $BE \parallel CD$, triangles ABE and ACD are similar.
Also

$$CD = 1.5BE$$

so

$$\frac{BE}{CD} = \frac{1}{1.5} = \frac{2}{3}$$

Therefore

$$\frac{AE}{AD} = \frac{2}{3}$$

Given $AE = 10$,

$$\frac{10}{AD} = \frac{2}{3}$$

$$AD = 15$$

So

$$ED = AD - AE = 15 - 10 = 5$$

For part (b),

$$AB = 2x + 5, \qquad BC = 3x - 5$$

$$AC = AB + BC = 5x$$

Using similarity,

$$\frac{AB}{AC} = \frac{2}{3}$$

$$\frac{2x+5}{5x} = \frac{2}{3}$$

$$3(2x + 5) = 10x$$

$$6x + 15 = 10x$$

$$x = 3.75$$

FINAL ANSWER
(a) $ED = 5$ cm, (b) $x = 3.75$.

PAST PAPER QUESTION Q#: 13 Marks: 3

TOPIC: **Circles, Arcs & Sectors**

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13 *OAB* is a sector of a circle with centre O and radius r cm.

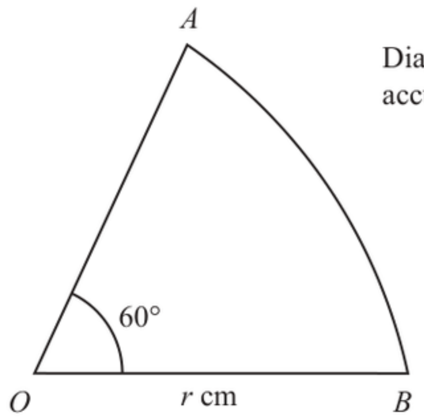

Diagram **NOT** accurately drawn

Angle $AOB = 60°$
The perimeter of the sector is P cm.

Find a formula for P in terms of r
Give your answer in the form $P = r(c\pi + k)$ where c and k are values to be found.

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION Q#: 13 Marks: 3

TOPIC: **Circles, Arcs & Sectors**

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WORKED SOLUTION

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Topic check. Circles, Arcs & Sectors. The tag is correct.

Solution

The arc length is

$$\frac{60}{360} \times 2\pi r = \frac{\pi r}{3}$$

The perimeter is two radii plus the arc:

$$P = 2r + \frac{\pi r}{3}$$

$$P = r\left(\frac{\pi}{3} + 2\right)$$

FINAL ANSWER

$P = r\left(\frac{1}{3}\pi + 2\right)$.

PAST PAPER QUESTION Q#: 14 Marks: 3

TOPIC: **Probability Toolkit**

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14 Adriana is going to roll a biased dice and spin a biased coin.

The probability that the coin will land on Heads is 0.8
The probability that the dice will land on 6 and the coin will land on Heads is 0.24

Work out the probability that the dice will land on 6 and the coin will land on Tails.

(Total for Question 14 is 3 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 3

TOPIC: **Probability Toolkit**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Toolkit. The tag is correct.

Solution

$$P(H) = 0.8$$

Given

$$P(6 \text{ and H}) = 0.24$$

$$P(6) = 0.24 \div 0.8 = 0.3$$

Also

$$P(T) = 1 - 0.8 = 0.2$$

$$P(6 \text{ and T}) = 0.3 \times 0.2 = 0.06$$

FINAL ANSWER

0.06.

PAST PAPER QUESTION Q#: 15 Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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15

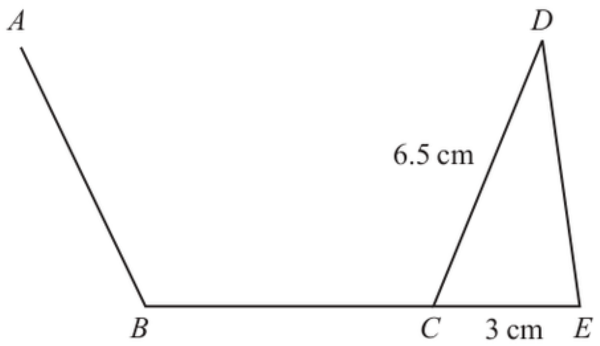

Diagram **NOT** accurately drawn

AB, *BC* and *CD* are three sides of a regular pentagon and *CDE* is a triangle.
BCE is a straight line.

$CD = 6.5\,\text{cm} \qquad CE = 3\,\text{cm}$

Work out the area of triangle *CDE*
Give your answer correct to 3 significant figures.

.... cm^2

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION

Q#: 15 Marks: 3

TOPIC: **Sine, Cosine Rule & Area of Triangles**

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WORKED SOLUTION

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Topic check. Corrected to sine/cosine rule and area of triangles. The key step is the included angle in triangle CDE.

Solution

A regular pentagon has interior angle

$$\frac{(5-2)\times 180}{5} = 108^\circ$$

Since BCE is a straight line,

$$\angle DCE = 180^\circ - 108^\circ = 72^\circ$$

Area of triangle CDE is

$$\frac{1}{2}ab\sin C$$

$$\text{Area} = \frac{1}{2}(6.5)(3)\sin 72^\circ$$

$$\text{Area} = 9.27\ldots$$

FINAL ANSWER

$9.27\ \text{cm}^2$ to 3 significant figures.

PAST PAPER QUESTION

Q#: 16 **Marks: 2**

TOPIC: **Graphs of Functions**

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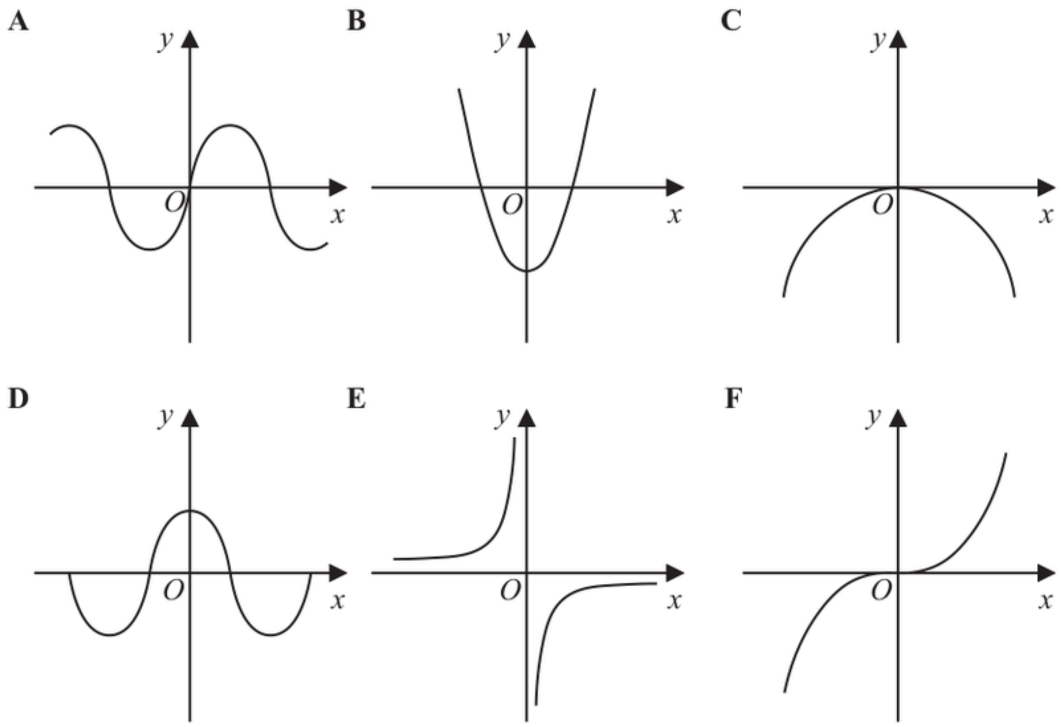
16 Here are six graphs.



Write down the letter of the graph that could have the equation

(i) $y = -\dfrac{1}{x}$

....

(1)

(ii) $y = \sin x°$

....

(1)

(Total for Question 16 is 2 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 2

TOPIC: **Graphs of Functions**

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WORKED SOLUTION

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Topic check. Corrected to Graphs of Functions.

Solution

(i)

$$y = -\frac{1}{x}$$

has branches in quadrants II and IV, so it is **Graph E**.

(ii)

$$y = \sin x^\circ$$

is the sine-shaped graph through the origin, so it is **Graph A**.

FINAL ANSWER

(i) E, (ii) A.

PAST PAPER QUESTION **Q#: 17** **Marks: 3**

TOPIC: **Functions**

Jun 2024 P2HR - Jun 2024 | P2HR

17 $\mathrm{f}(x) = \dfrac{x}{2x - 4}$ $\qquad$ $\mathrm{g}(x) = 3x + 1$

Given that $\mathrm{fg}(k) = 2$

work out the value of k

$k =$ ....

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17 Marks: 3

TOPIC: **Functions**

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WORKED SOLUTION

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Topic check. Functions. The tag is correct.

Solution

$$f(x) = \frac{k}{2x-4}, \qquad g(x) = 3x+1$$

Given

$$fg(k) = 2$$

This means

$$f(g(k)) = 2$$

Now

$$g(k) = 3k+1$$

So

$$f(g(k)) = \frac{k}{2(3k+1)-4} = \frac{k}{6k-2}$$

$$\frac{k}{6k-2} = 2$$

$$k = 12k - 4$$

$$11k = 4$$

$$k = \frac{4}{11}$$

FINAL ANSWER

$k = \frac{4}{11}$.

PAST PAPER QUESTION

Q#: 18 Marks: 2

TOPIC: **Algebraic Proof**

Jun 2024 P2HR - Jun 2024 | P2HR

18 Use algebra to show that $0.\dot{3}0\dot{6} = \dfrac{34}{111}$

(Total for Question 18 is 2 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 2

TOPIC: **Algebraic Proof**

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Topic check. Algebraic proof. The tag is correct.

Solution

Let

$$x = 0.\dot{3}0\dot{6}$$

Then

$$1000x = 306.\dot{3}0\dot{6}$$

Subtract:

$$1000x - x = 306$$

$$999x = 306$$

$$x = \frac{306}{999} = \frac{34}{111}$$

FINAL ANSWER

$0.\dot{3}0\dot{6} = \dfrac{34}{111}$.

PAST PAPER QUESTION Q#: 19 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jun 2024 P2HR - Jun 2024 | P2HR

19 Aviv goes on a cycle journey.

For the cycle journey

average speed = 19 km/h correct to the nearest whole number

time = 1.5 hours correct to one decimal place

Work out the upper bound for the distance Aviv travels.
Give your answer correct to 3 significant figures.

.... km

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Rounding, Estimation & Bounds**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

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Topic check. Rounding, Estimation and Bounds. The tag is correct.

Method

Distance is found using

$$\text{distance} = \text{speed} \times \text{time}$$

For the upper bound,

$$\text{speed} < 19.5, \quad \text{time} < 1.55$$

$$\text{distance}_{\text{upper}} = 19.5 \times 1.55 = 30.225$$

Correct to 3 significant figures,

$$30.225 \approx 30.2$$

FINAL ANSWER

30.2 km

PAST PAPER QUESTION Q#: 20 Marks: 4

TOPIC: **Solving Inequalities**

Jun 2024 P2HR - Jun 2024 | P2HR

20 Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Solving Inequalities**

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WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Solving Quadratic Equations to Solving Inequalities.

Method

$$6x^2 - 7x - 20 > 0$$

Factorise:

$$6x^2 - 7x - 20 = (3x+4)(2x-5)$$

The critical values are

$$x = -\frac{4}{3} \quad \text{and} \quad x = \frac{5}{2}$$

Since the coefficient of x^2 is positive, the quadratic is greater than 0 outside the roots.

FINAL ANSWER

$x < -\frac{4}{3}$ or $x > \frac{5}{2}$

PAST PAPER QUESTION Q#: 20 Marks: 4

TOPIC: **Solving Inequalities**

Jun 2024 P2HR - Jun 2024 | P2HR

20 Solve $6x^2 - 7x - 20 > 0$
Show clear algebraic working.

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 4

TOPIC: **Solving Inequalities**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Solving Quadratic Equations to Solving Inequalities.

Method

$$6x^2 - 7x - 20 > 0$$

Factorise:

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The critical values are

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Since the coefficient of x^2 is positive, the quadratic is greater than 0 outside the roots.

FINAL ANSWER

$x < -\frac{4}{3}$ or $x > \frac{5}{2}$

PAST PAPER QUESTION

TOPIC: **Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

Q#: 21 Marks: 4

21 *ABCD* is a square.

The point *A* has coordinates $(-5, 2)$
The point *B* has coordinates $(3, 5)$

Find an equation of the line that passes through *B* and *C*
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$A = (-5, 2), \qquad B = (3, 5)$$

The gradient of AB is

$$\frac{5-2}{3-(-5)} = \frac{3}{8}$$

Since $ABCD$ is a square, BC is perpendicular to AB.
So the gradient of BC is

$$-\frac{8}{3}$$

The line through $B = (3, 5)$ is

$$y - 5 = -\frac{8}{3}(x-3)$$

Multiply by 3:

$$3y - 15 = -8x + 24$$

$$8x + 3y - 39 = 0$$

FINAL ANSWER

$8x + 3y - 39 = 0$

PAST PAPER QUESTION Q#: 21 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

21 *ABCD* is a square.

The point A has coordinates $(-5, 2)$
The point B has coordinates $(3, 5)$

Find an equation of the line that passes through B and C
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION

Q#: 21 Marks: 4

TOPIC: **Coordinate Geometry**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$A = (-5, 2), \qquad B = (3, 5)$$

The gradient of AB is

$$\frac{5-2}{3-(-5)} = \frac{3}{8}$$

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Multiply by 3:

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$$8x + 3y - 39 = 0$$

FINAL ANSWER

$8x + 3y - 39 = 0$

PAST PAPER QUESTION

Q#: 22 **Marks: 5**

TOPIC: **Simultaneous Equations**

Jun 2024 P2HR - Jun 2024 | P2HR

22 Solve the simultaneous equations

$$\begin{aligned} x^2 + y^2 &= y + 11 \\ y &= 3x - 1 \end{aligned}$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Substitute:

$$x^2 + (3x-1)^2 = 3x - 1 + 11$$

$$x^2 + 9x^2 - 6x + 1 = 3x + 10$$

$$10x^2 - 9x - 9 = 0$$

$$(5x+3)(2x-3) = 0$$

So

$$x = -\frac{3}{5} \quad \text{or} \quad x = \frac{3}{2}$$

Find y:

$$x = -\frac{3}{5} \Rightarrow y = -\frac{14}{5}$$

$$x = \frac{3}{2} \Rightarrow y = \frac{7}{2}$$

FINAL ANSWER

$(x, y) = \left(-\frac{3}{5}, -\frac{14}{5}\right)$ or $\left(\frac{3}{2}, \frac{7}{2}\right)$

PAST PAPER QUESTION Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2024 P2HR - Jun 2024 | P2HR

22 Solve the simultaneous equations

$$x^2 + y^2 = y + 11$$
$$y = 3x - 1$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$x^2 + y^2 = y + 11$$

$$y = 3x - 1$$

Substitute:

$$x^2 + (3x-1)^2 = 3x - 1 + 11$$

$$x^2 + 9x^2 - 6x + 1 = 3x + 10$$

$$10x^2 - 9x - 9 = 0$$

$$(5x+3)(2x-3) = 0$$

So

$$x = -\frac{3}{5} \quad \text{or} \quad x = \frac{3}{2}$$

Find y:

$$x = -\frac{3}{5} \Rightarrow y = -\frac{14}{5}$$

$$x = \frac{3}{2} \Rightarrow y = \frac{7}{2}$$

FINAL ANSWER

$(x, y) = \left(-\frac{3}{5}, -\frac{14}{5}\right)$ or $\left(\frac{3}{2}, \frac{7}{2}\right)$

PAST PAPER QUESTION

Topic: **Transformations of Graphs**

Jun 2024 P2HR - Jun 2024 | P2HR

Q#: 23 Marks: 2

23 A curve has equation $y = f(x)$
The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 10$

(.............................. , ...)
(1)

(ii) $y = f(3x)$

(.............................. , ...)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The minimum point of $y = f(x)$ is $(6, -3)$.
For

$$y = f(x) + 10$$

the graph moves 10 units up:

$$(6, -3) \to (6, 7)$$

For

$$y = f(3x)$$

the x-coordinate is divided by 3:

$$(6, -3) \to (2, -3)$$

FINAL ANSWER
$(6, 7)$ and $(2, -3)$

PAST PAPER QUESTION Q#: 23 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2024 P2HR - Jun 2024 | P2HR

23 A curve has equation $y = f(x)$
The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

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(.............................. , ...)
(1)

(ii) $y = f(3x)$

(.............................. , ...)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The minimum point of $y = f(x)$ is $(6, -3)$.
For

$$y = f(x) + 10$$

the graph moves 10 units up:

$$(6, -3) \to (6, 7)$$

For

$$y = f(3x)$$

the x-coordinate is divided by 3:

$$(6, -3) \to (2, -3)$$

FINAL ANSWER
$(6, 7)$ and $(2, -3)$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jun 2024 P2HR - Jun 2024 | P2HR

24 The diagram shows a solid, **S**, made from a cone and a hemisphere.

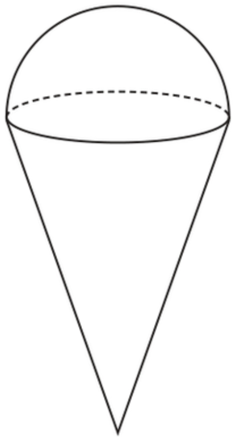

Diagram **NOT** accurately drawn

The centre of the circular face of the cone coincides with the centre of the flat surface of the hemisphere.

The radius of the circular face of the cone, x cm, is equal to the radius of the hemisphere.

The total height of **S** is 4 × the radius of the hemisphere.

A separate sphere has radius kx cm.
The volume of this sphere is 12.5 × the volume of **S**

(a) Work out the value of k

k = ...

(4)

A solid, **T**, is similar to solid **S**

The volume of **T** is 512 × the volume of **S**

The total surface area of **T** is d × the total surface area of **S**

(b) Find the value of d

d = ...

(1)

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The hemisphere has radius x, so its height is x.
The total height of solid S is $4x$, so the cone height is

$$4x - x = 3x$$

Volume of the cone:

$$\frac{1}{3}\pi x^2(3x) = \pi x^3$$

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

So the volume of solid S is

$$\pi x^3 + \frac{2}{3}\pi x^3 = \frac{5}{3}\pi x^3$$

The sphere has radius kx, so its volume is

$$\frac{4}{3}\pi (kx)^3 = \frac{4}{3}\pi k^3 x^3$$

This is 12.5 times the volume of S:

$$\frac{4}{3}\pi k^3 x^3 = 12.5\left(\frac{5}{3}\pi x^3\right)$$

Cancel $\frac{1}{3}\pi x^3$:

$$4k^3 = 62.5$$

$$k^3 = 15.625 = \frac{125}{8}$$

$$k = 2.5$$

For solid T, the volume scale factor is 512.

$$512 = 8^3$$

So the linear scale factor is 8, and the surface area scale factor is

$$8^2 = 64$$

FINAL ANSWER

$\boxed{k = 2.5,\ d = 64}$

PAST PAPER QUESTION

Q#: 24 Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jun 2024 P2HR - Jun 2024 | P2HR

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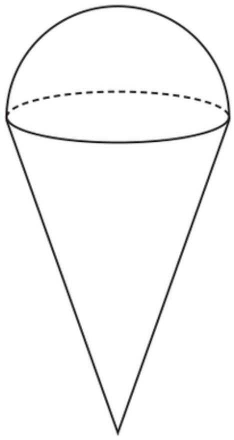

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The total surface area of **T** is d × the total surface area of **S**

(b) Find the value of d

d = ...

(1)

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24 **Marks: 5**

TOPIC: **Area & Volume of Similar Shapes**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

$\boxed{k = 2.5,\ d = 64}$

PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

25 *OPQR* is a parallelogram.

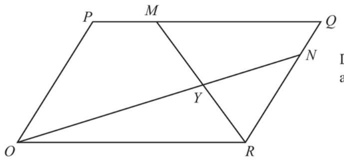

Diagram **NOT** accurately drawn

$\overrightarrow{OP} = 2\mathbf{a}$ and $\overrightarrow{OR} = 3\mathbf{b}$

The point M lies on PQ such that $PM = \frac{1}{4}PQ$

The point N lies on RQ such that $RN = \frac{4}{5}RQ$

(a) Find, in terms of **a** and **b**, giving your answers in simplest form

(i) $\overrightarrow{ON}$

....

(1)

(ii) $\overrightarrow{MR}$

....

(1)

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

$k =$ ....

(4)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION Q#: 25 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$\overrightarrow{OP} = 2a, \qquad \overrightarrow{OR} = 3b$$

So

$$\overrightarrow{OQ} = 2a + 3b$$

Since $PM : PQ = 1 : 4$,

$$\overrightarrow{PM} = \frac{1}{4}\overrightarrow{PQ}$$

$$\overrightarrow{PM} = \frac{1}{4}(3b) = \frac{3}{4}b$$

$$\overrightarrow{OM} = 2a + \frac{3}{4}b$$

Since $RN : RQ = 4 : 5$,

$$\overrightarrow{RN} = \frac{4}{5}\overrightarrow{RQ}$$

$$\overrightarrow{RN} = \frac{4}{5}(2a) = \frac{8}{5}a$$

$$\overrightarrow{ON} = 3b + \frac{8}{5}a$$

So

$$\overrightarrow{MR} = \overrightarrow{OR} - \overrightarrow{OM}$$

$$\overrightarrow{MR} = 3b - \left(2a + \frac{3}{4}b\right)$$

$$\overrightarrow{MR} = -2a + \frac{9}{4}b$$

Let Y be on ON, so

$$\overrightarrow{OY} = k\overrightarrow{ON} = k\left(\frac{8}{5}a + 3b\right)$$

Also Y is on MR, so

$$\overrightarrow{OY} = \overrightarrow{OM} + t\overrightarrow{MR}$$

$$\overrightarrow{OY} = 2a + \frac{3}{4}b + t\left(-2a + \frac{9}{4}b\right)$$

Compare coefficients of a and b:

$$\frac{8}{5}k = 2 - 2t$$

$$3k = \frac{3}{4} + \frac{9}{4}t$$

From the first equation,

$$t = 1 - \frac{4}{5}k$$

Substitute into the second equation:

$$3k = \frac{3}{4} + \frac{9}{4}\left(1 - \frac{4}{5}k\right)$$

$$3k = 3 - \frac{9}{5}k$$

$$\frac{24}{5}k = 3$$

$$k = \frac{5}{8}$$

FINAL ANSWER
$\overrightarrow{ON} = \frac{8}{5}a + 3b$, $\overrightarrow{MR} = -2a + \frac{9}{4}b$, and $k = \frac{5}{8}$

PAST PAPER QUESTION

Q#: 25 Marks: 6

TOPIC: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

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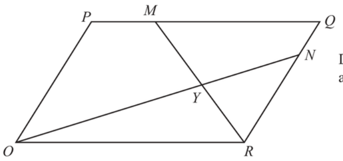

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The point N lies on RQ such that $RN = \frac{4}{5}RQ$

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(i) $\overrightarrow{ON}$

....
(1)

(ii) $\overrightarrow{MR}$

....
(1)

MR and ON intersect at the point Y

Given that

$$OY = k \times ON$$

(b) use a vector method to find the value of k

$k =$ ....
(4)

(Total for Question 25 is 6 marks)

PAST PAPER SOLUTION Q#: 25 Marks: 6

Topic: **Vectors**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

$$\overrightarrow{OP} = 2a, \qquad \overrightarrow{OR} = 3b$$

So

$$\overrightarrow{OQ} = 2a + 3b$$

Since $PM : PQ = 1 : 4$,

$$\overrightarrow{PM} = \frac{1}{4}\overrightarrow{PQ}$$

$$\overrightarrow{PM} = \frac{1}{4}(3b) = \frac{3}{4}b$$

$$\overrightarrow{OM} = 2a + \frac{3}{4}b$$

Since $RN : RQ = 4 : 5$,

$$\overrightarrow{RN} = \frac{4}{5}\overrightarrow{RQ}$$

$$\overrightarrow{RN} = \frac{4}{5}(2a) = \frac{8}{5}a$$

$$\overrightarrow{ON} = 3b + \frac{8}{5}a$$

So

$$\overrightarrow{MR} = \overrightarrow{OR} - \overrightarrow{OM}$$

$$\overrightarrow{MR} = 3b - \left(2a + \frac{3}{4}b\right)$$

$$\overrightarrow{MR} = -2a + \frac{9}{4}b$$

Let Y be on ON, so

$$\overrightarrow{OY} = k\overrightarrow{ON} = k\left(\frac{8}{5}a + 3b\right)$$

Also Y is on MR, so

$$\overrightarrow{OY} = \overrightarrow{OM} + t\overrightarrow{MR}$$

$$\overrightarrow{OY} = 2a + \frac{3}{4}b + t\left(-2a + \frac{9}{4}b\right)$$

Compare coefficients of a and b:

$$\frac{8}{5}k = 2 - 2t$$

$$3k = \frac{3}{4} + \frac{9}{4}t$$

From the first equation,

$$t = 1 - \frac{4}{5}k$$

Substitute into the second equation:

$$3k = \frac{3}{4} + \frac{9}{4}\left(1 - \frac{4}{5}k\right)$$

$$3k = 3 - \frac{9}{5}k$$

$$\frac{24}{5}k = 3$$

$$k = \frac{5}{8}$$

FINAL ANSWER
$\overrightarrow{ON} = \frac{8}{5}a + 3b$, $\overrightarrow{MR} = -2a + \frac{9}{4}b$, and $k = \frac{5}{8}$

PAST PAPER QUESTION

TOPIC: **Algebraic Fractions**

Jun 2024 P2HR - Jun 2024 | P2HR

Q#: 26 Marks: 4

26 Write $4 - \left[(3x-5) \div \dfrac{3x^2+x-10}{4x-1}\right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$4 - \left[(3x-5) \div \frac{3x^2+x-10}{4x-1}\right]$$

Factor the quadratic:

$$3x^2 + x - 10 = (3x-5)(x+2)$$

So the expression inside the square brackets is

$$(3x-5) \div \frac{(3x-5)(x+2)}{4x-1}$$

Dividing by a fraction means multiplying by its reciprocal:

$$(3x-5) \times \frac{4x-1}{(3x-5)(x+2)}$$

Cancel $3x-5$:

$$\frac{4x-1}{x+2}$$

Therefore the full expression is

$$4 - \frac{4x-1}{x+2}$$

Use the common denominator $x+2$:

$$\frac{4(x+2)-(4x-1)}{x+2}$$

$$= \frac{4x+8-4x+1}{x+2}$$

$$= \frac{9}{x+2}$$

FINAL ANSWER

$$\boxed{\frac{9}{x+2}}$$

PAST PAPER QUESTION Q#: 26 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2024 P2HR - Jun 2024 | P2HR

26 Write $4 - \left[(3x-5) \div \dfrac{3x^2 + x - 10}{4x-1}\right]$ as a single fraction in its simplest form.

(Total for Question 26 is 4 marks)

PAST PAPER SOLUTION

Q#: 26 Marks: 4

TOPIC: **Algebraic Fractions**

Jun 2024 P2HR - Jun 2024 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Sine/Cosine Rule to Algebraic Fractions.

Method

$$4 - \left[(3x-5) \div \frac{3x^2+x-10}{4x-1}\right]$$

Factor the quadratic:

$$3x^2 + x - 10 = (3x-5)(x+2)$$

So the expression inside the square brackets is

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Dividing by a fraction means multiplying by its reciprocal:

$$(3x-5) \times \frac{4x-1}{(3x-5)(x+2)}$$

Cancel $3x-5$:

$$\frac{4x-1}{x+2}$$

Therefore the full expression is

$$4 - \frac{4x-1}{x+2}$$

Use the common denominator $x+2$:

$$\frac{4(x+2)-(4x-1)}{x+2}$$

$$= \frac{4x+8-4x+1}{x+2}$$

$$= \frac{9}{x+2}$$

FINAL ANSWER

$$\boxed{\frac{9}{x+2}}$$